24/07/2026

lec-1 (Introductory class)

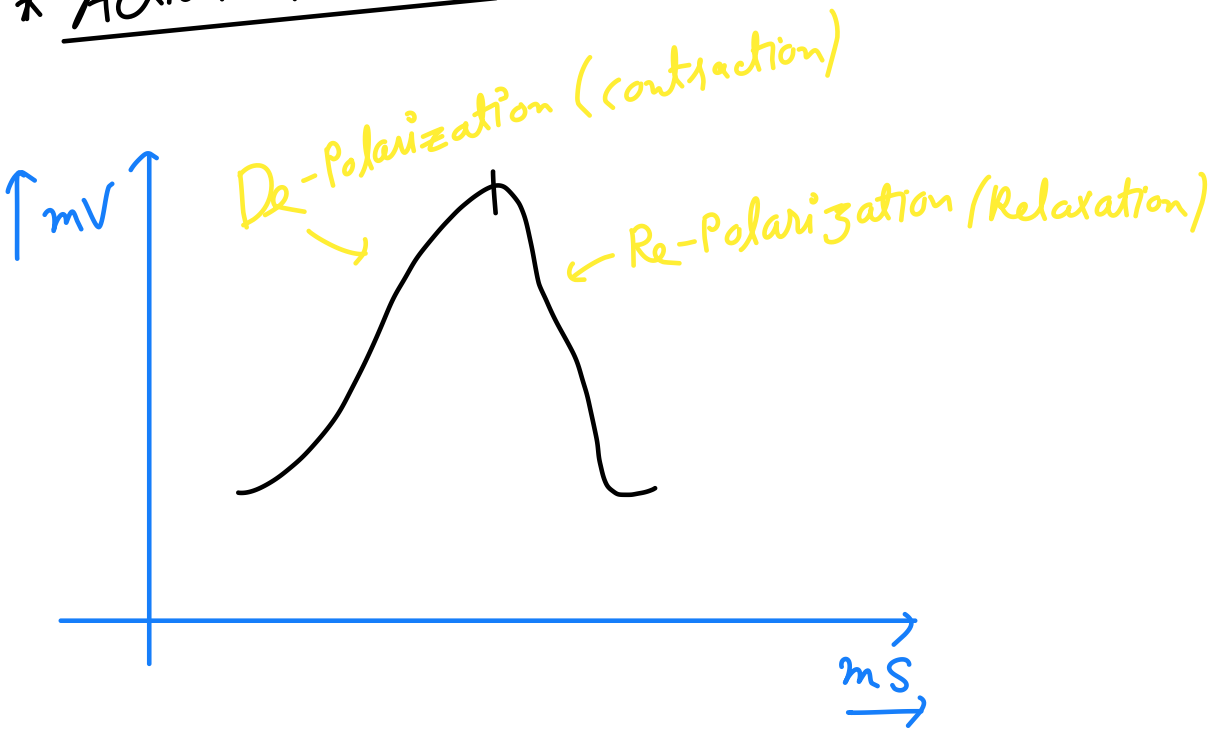
27/07/26 lec-2

* Human Body as a System

1. Skeletal System.
2. Circulatory System.
3. Respiratory System.
4. Digestive System.
5. Excretory System.
6. Regulatory System.
7. Reproductive System.
8. Muscular System.

* Action Potential

Q: Define Action-Potential



During the De & Re polarization of a cell-membrane, a potential develops in the cell membrane, it is called Action Potential.

* Characteristics of Action Potential:

① Action potential generated due to Ionic Voltage

② It has delayed transition in comparison with electronic circuit

③ The Action Potential give rise to a Voltage waveform.

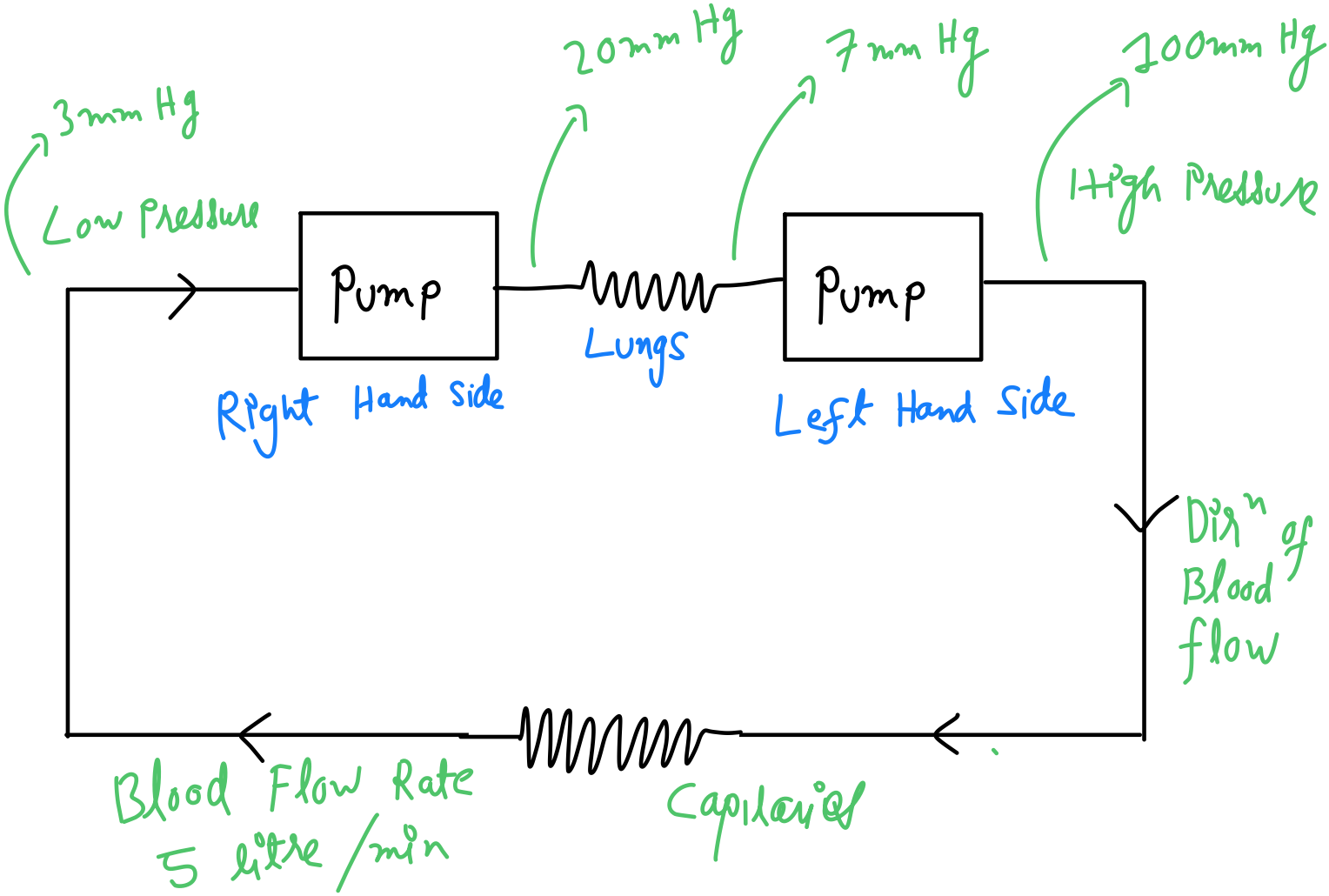
④ The coordinated electrochemical activities of large number of cells give rise to Bio-potential. Ex: ECG, EEG, EMG

* Propagation of Action Potential

① Muscle Fiber or Unmyelinated nerve fibers.

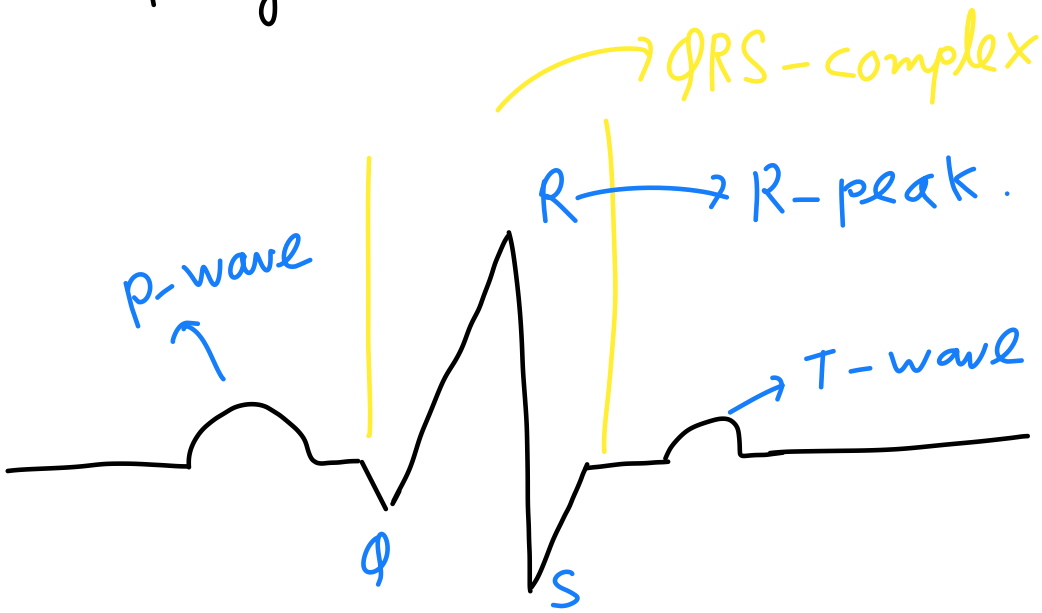
② Myelinated nerve.

• Cardiovascular System:



* Electrocardiogram (ECG):

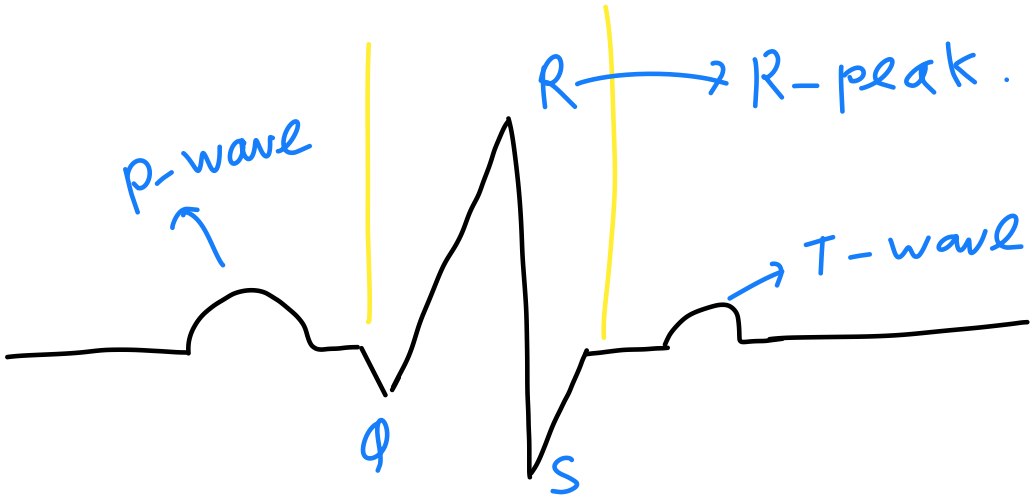
- Amplitude → mV → 0.1 to 5 mV
- Frequency → 0.01 to 100 Hz



Normal ECG during one cycle

ECG:
Also known as Quasi-periodic signal.

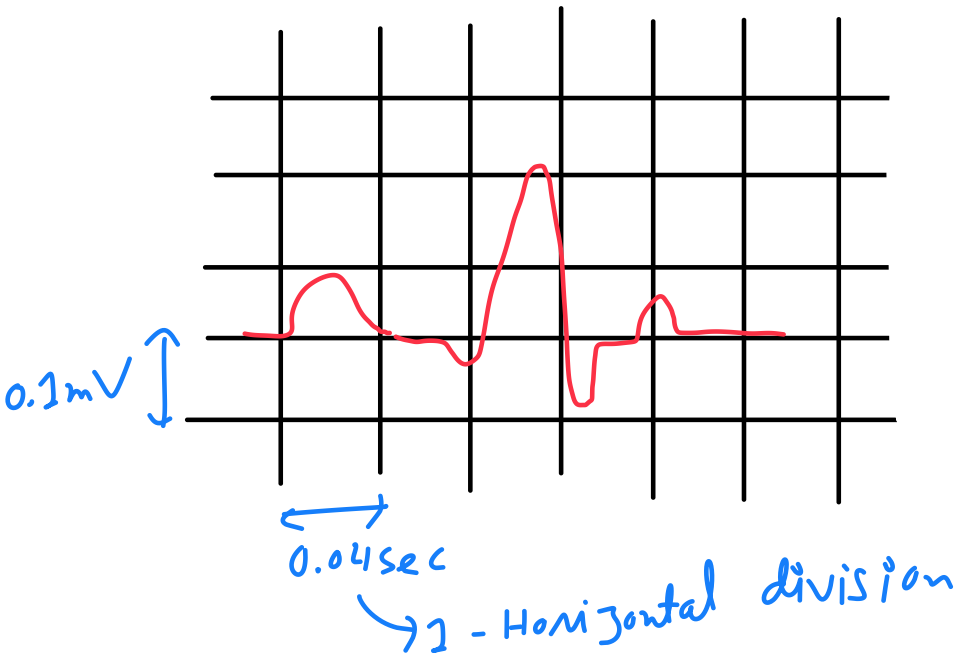
lec-3 (30/07/26)



* P-wave: generates due to contraction of Atrium.

* QRS-complex: generates due to contraction of ventricles.

* T-wave: generates due to depolarization of Ventricles.



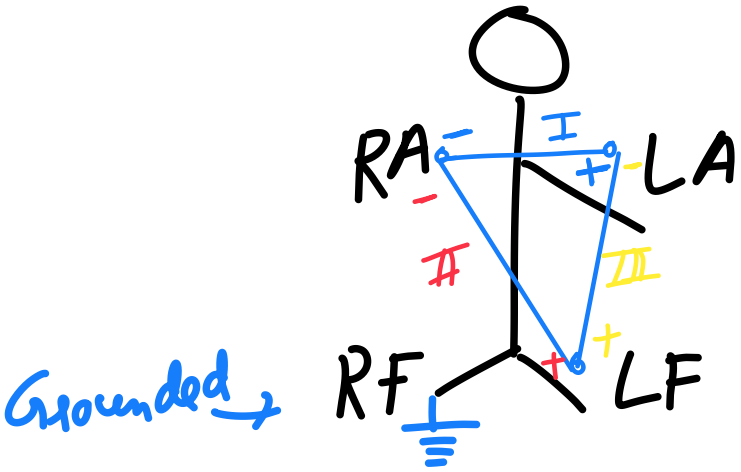
* Acquisition of ECG:

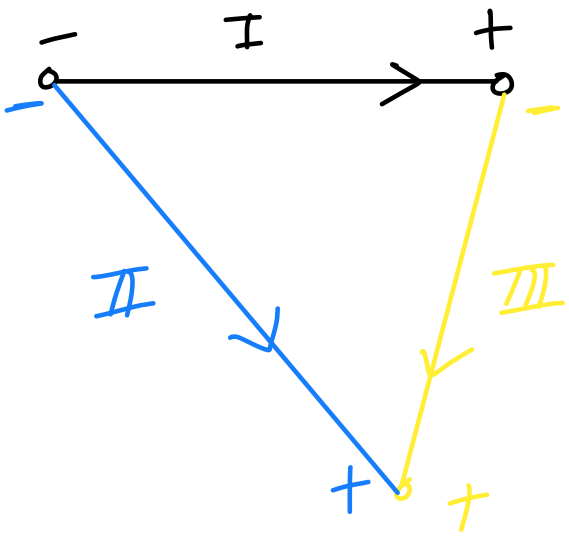
① Bipolar lead configuration:

Lead-I

Lead-II

Lead-III





Einthoven Triangle Rule:

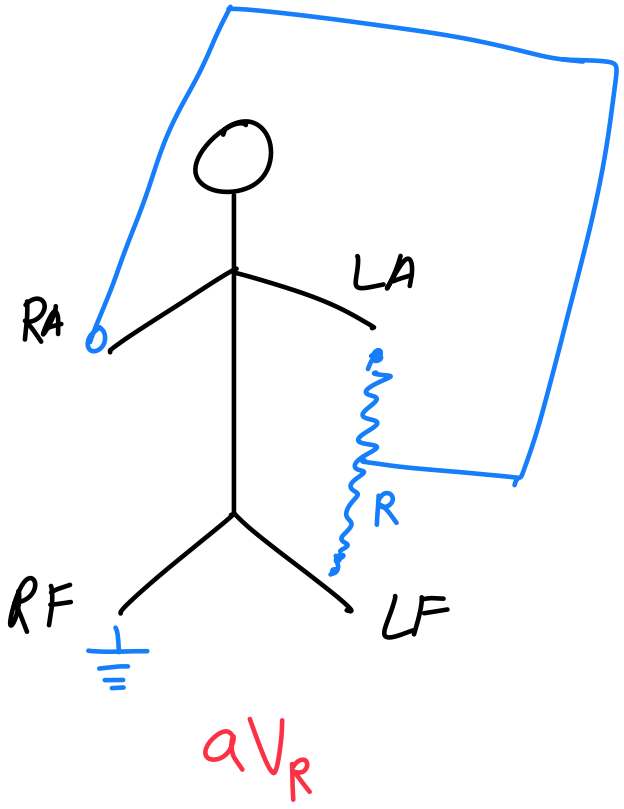
Lead II = Lead I + Lead III

② Augmented Lead Configuration: (Unipolar Lead configuration)

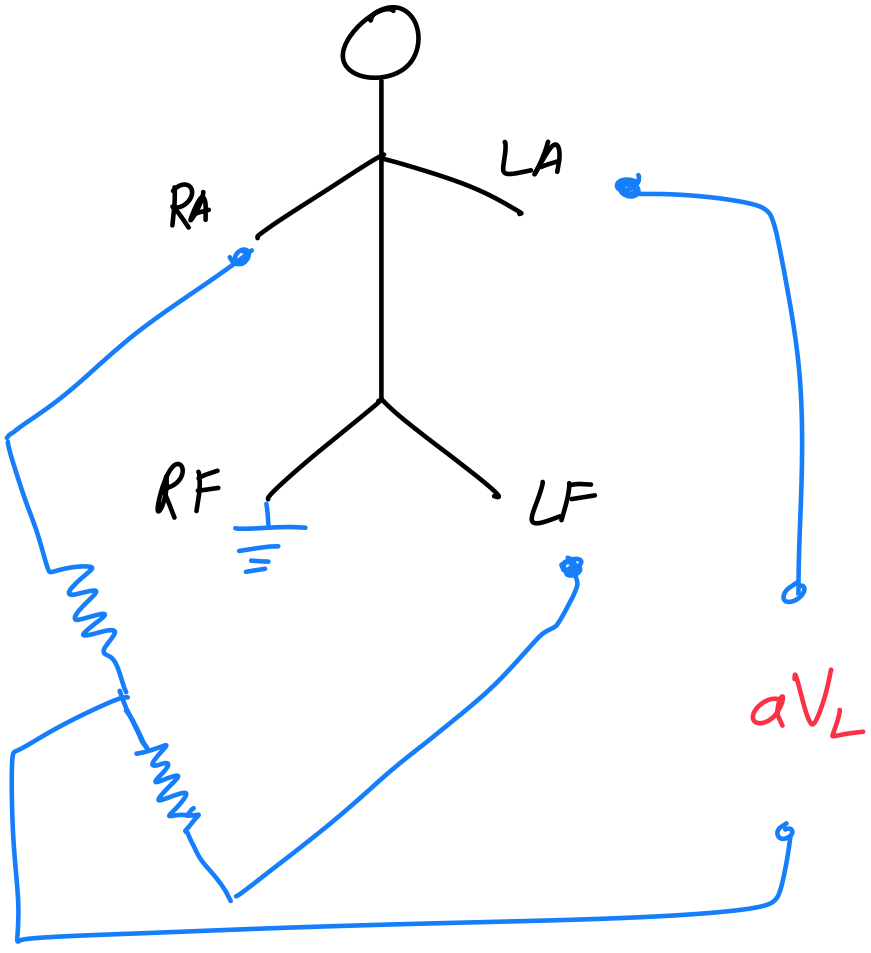
$$aV_R \rightarrow \text{Right Arm} = \frac{I + II}{2}$$

$$aV_L \rightarrow \text{Left Arm} = \frac{I - III}{2}$$

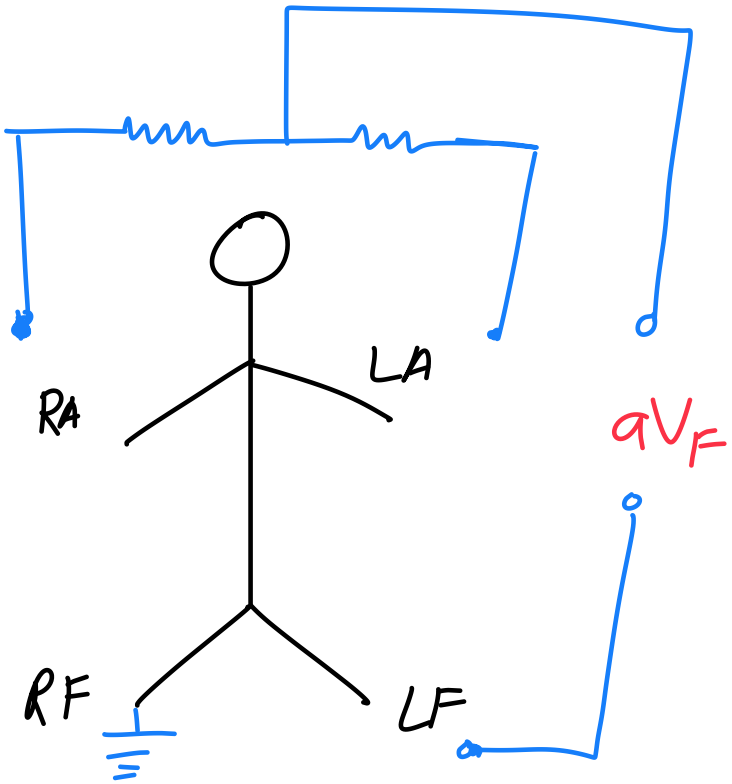
$$aV_F \rightarrow \text{Foot (left)} = \frac{II + III}{2}$$



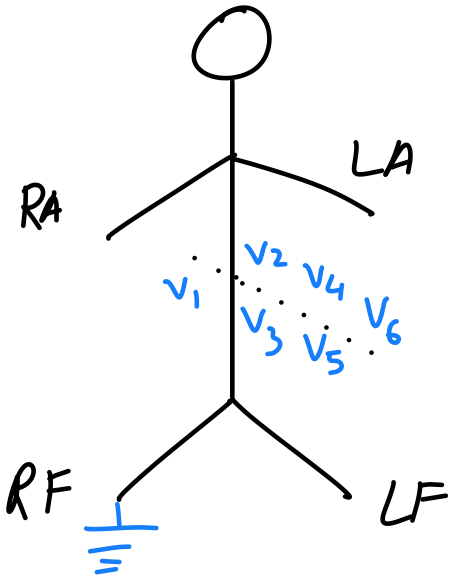
aV_R

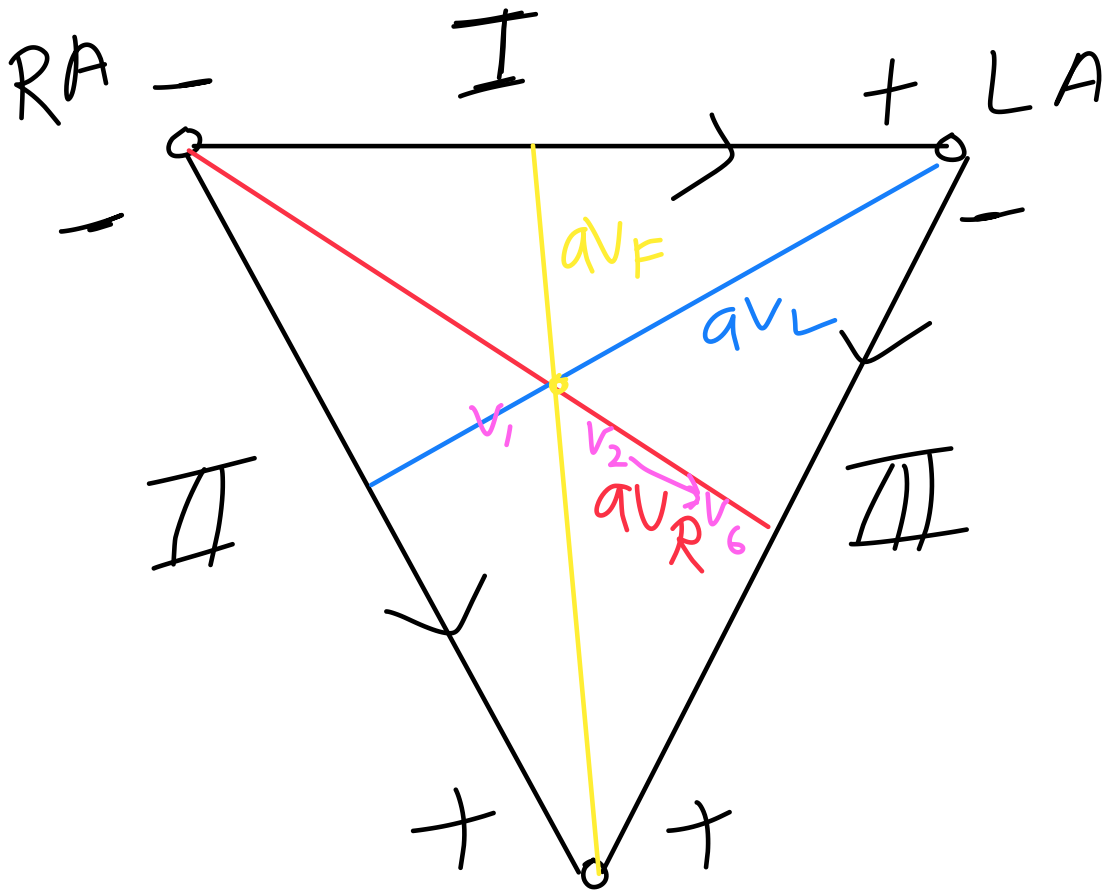


aV_L



(3) Chest Leads configuration (Unipolar lead configuration)





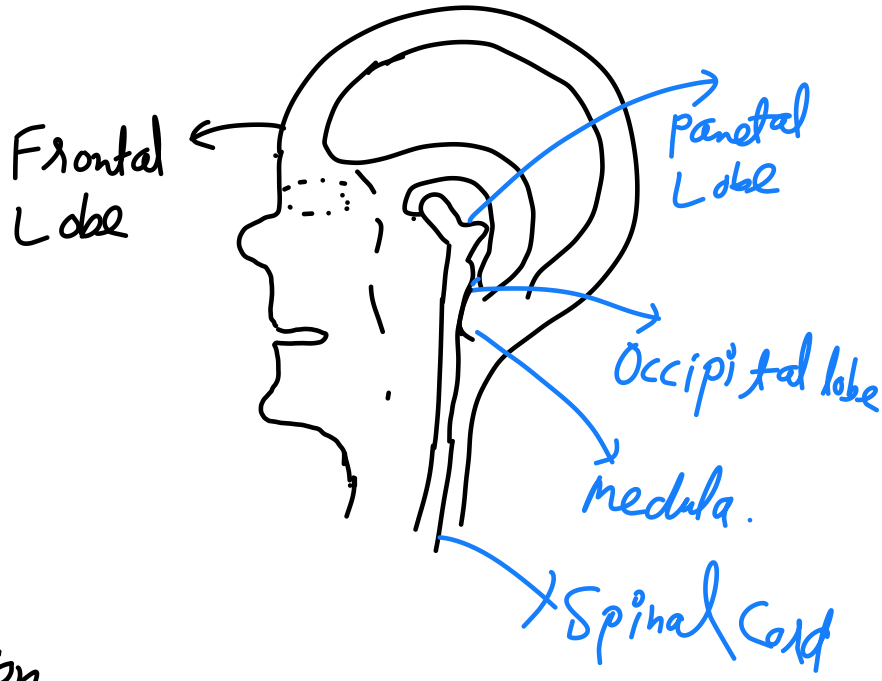
* Heart Rate : → beats/min

- HR < 60 bpm
  - ↳ BradyCardia
- HR > 100 bpm
  - ↳ TachyCardia

Lec- 4 (03/08/2026)

* Electroencephalogram (EEG)

* Frontal lobe is responsible for intelligence



* Occipital lobe is responsible for Vision.
* Parietal lobe is responsible for sensation
* Medula is responsible for autonomous function such as Breathing, contraction and regulation of the heart.

- Brain → Neuron
  - → Sensory → Sensing
  - → Interneuron → Central Nervous system (Spinal cord, Brain)
  - → Motor

* Sensory neurons provide Sensing.

* Interneurons are the neurons of the Central Nervous system. Interneurons provide Decision-making.

* Motor neurons provide Autonomous action.

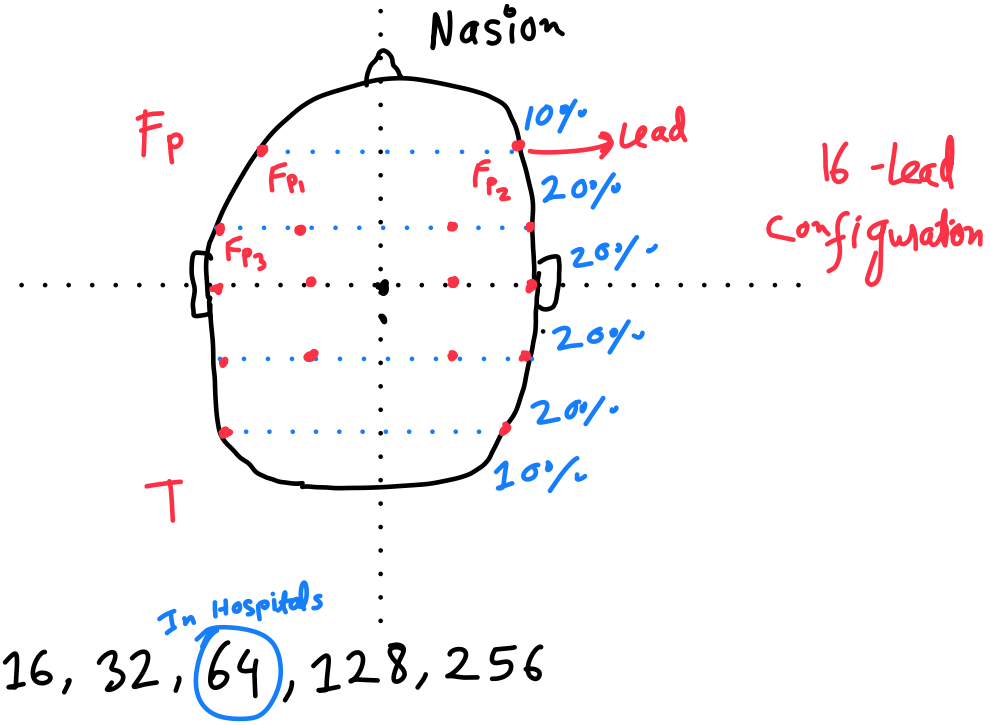
* EEG recording Configuration

① Unipolar EEG

② Bipolar "

③ Average. "

* EEG Lead positions



In Hospitals

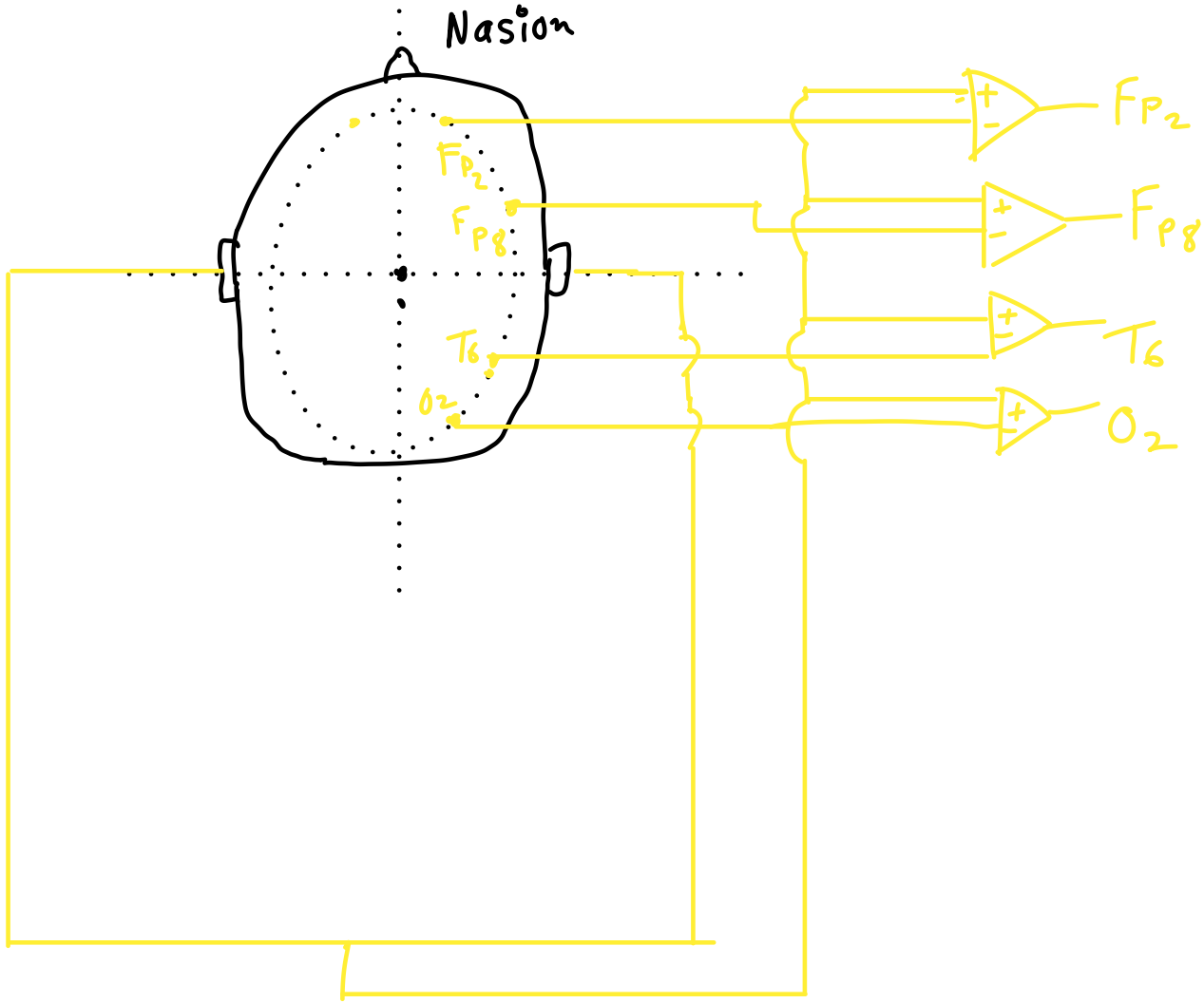
16, 32, (64), 128, 256

Lec-5 (06/08/2026)

EEG Recording Configuration

(1) Unipolar EEG Recording Configuration.

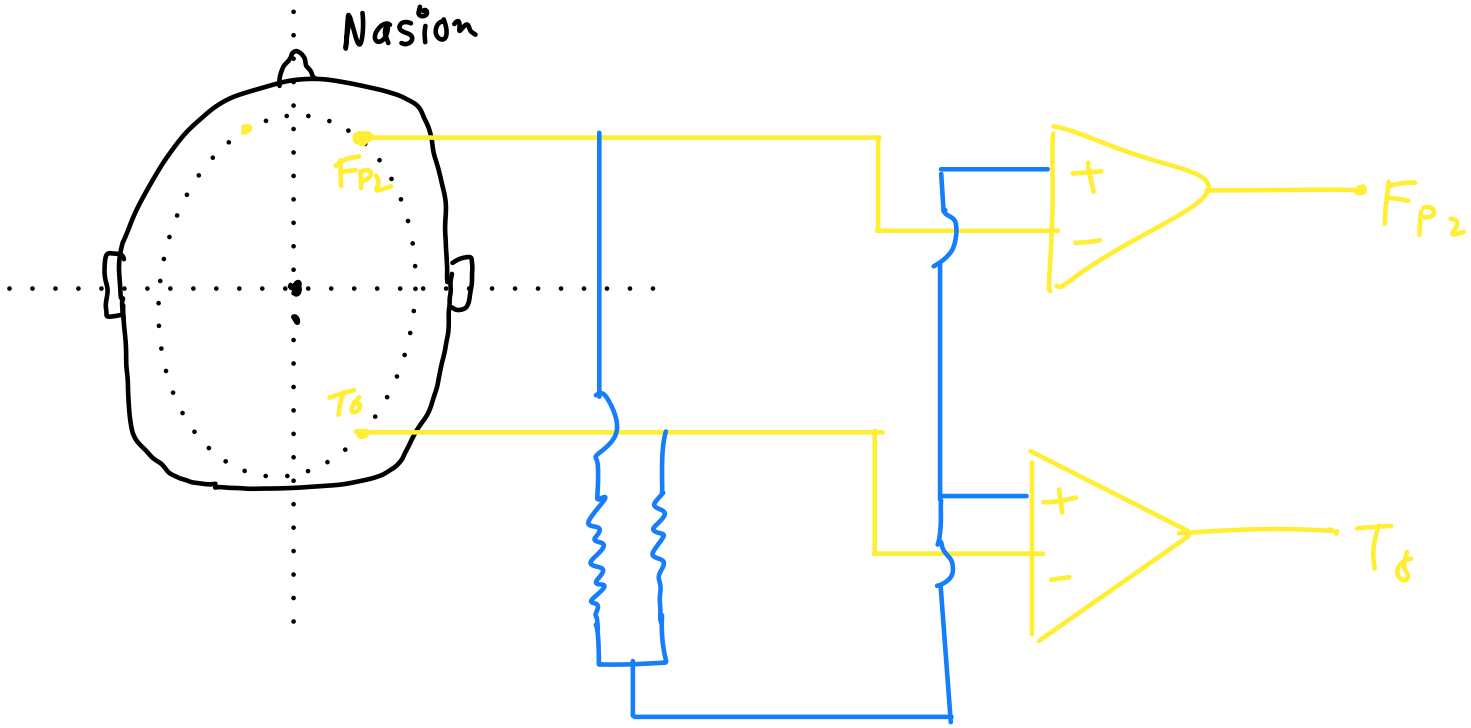
→ To know if problem is present or not



* In the Unipolar EEG Recording Configuration, the reference point is mid-potential of two Ear leads
* Unipolar EEG Recording Configuration. provides information about Global activity

of our brain. It indicates which part of our brain is more active during different activities.

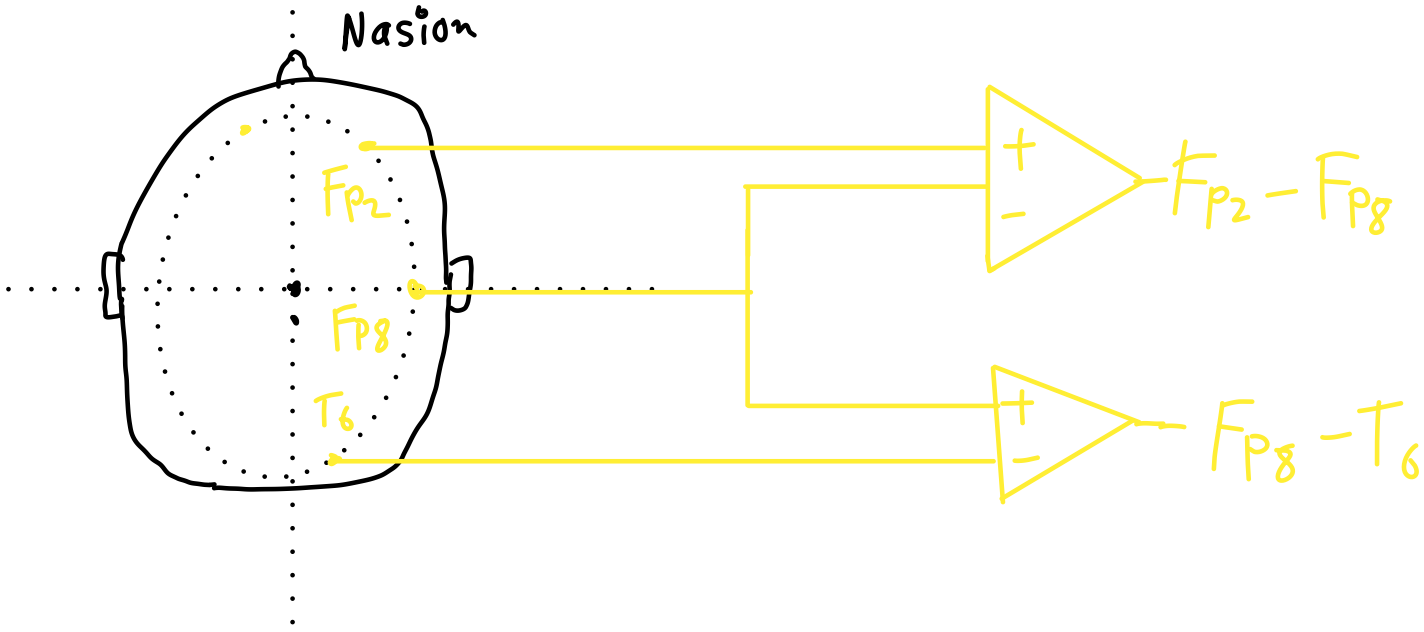
(2) Average EEG Recording Configuration



* In Average EEG recording configuration, the reference point is average of all the leads

* The nature of EEG signal is similar to Unipolar EEG Recording Configuration (Global) → No localised information.

③ Bipolar EEG Recording Configuration



* Bipolar EEG Recording Configuration provides Localized information of our brain, it is the reason that it is widely used for diagnostic purposes.

EEG: • Amplitude → μV

• Frequency → 0.01 to 100 Hz

Lec-6 (07-08-2026)

* EEG signal frequency bands

Name	Frequency Range	Activity	Abnormality	Signal waveform
1. Delta-wave	0.5 - 4.0 Hz	Deep Sleep	Lesions	
2. Theta -wave	4.0 - 8.0 Hz	Sleep	Depression	
3. Alpha - wave	8.0 - 13.0 Hz	Awake at Relaxation		
4. Beta - wave	13.0 - 22.0 Hz	Awake at Activity		
5. Gamma - wave	22.0 - 33.0 Hz	Higher mental Activity		

- Sudden spike in EEG waveform → Seizure

* Applications of EEG Signals:

① It is used to determine Anasthetic level

② For Epilepsy detection. → Periodic spikes in EEG

③ Brain injury. → By physical damage or Trauma

④ Tumor detection & localization.

⑤ Alertness monitoring.

⑥ Monitoring during Surgery.

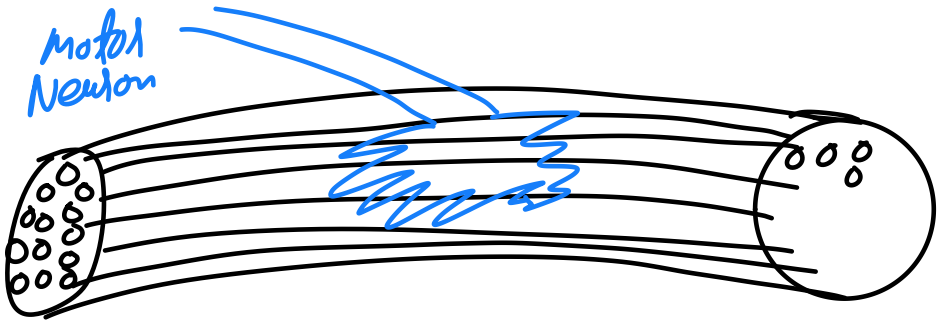
⑦ Efficacy of Yoga.

* Electromyogram (EMG):

* The recording of electrical stimulations generated by our muscles is called Electromyogram and the process is called Electromyography.

Lec-7 (10/08/2026)

* Electromyogram (EMG)



* Motor Neuron terminates on Muscle Fibers
  - A single motor neuron terminates not on a single muscle fiber on a connection of Muscle fiber which is called Motor unit.
  - And when motor neuron stimulates muscles contraction and relaxation happens.

This phenomenon is called Twitch.

- Potential generated by a connection of muscle fibers is called Motor unit Action Potential (MUAP)

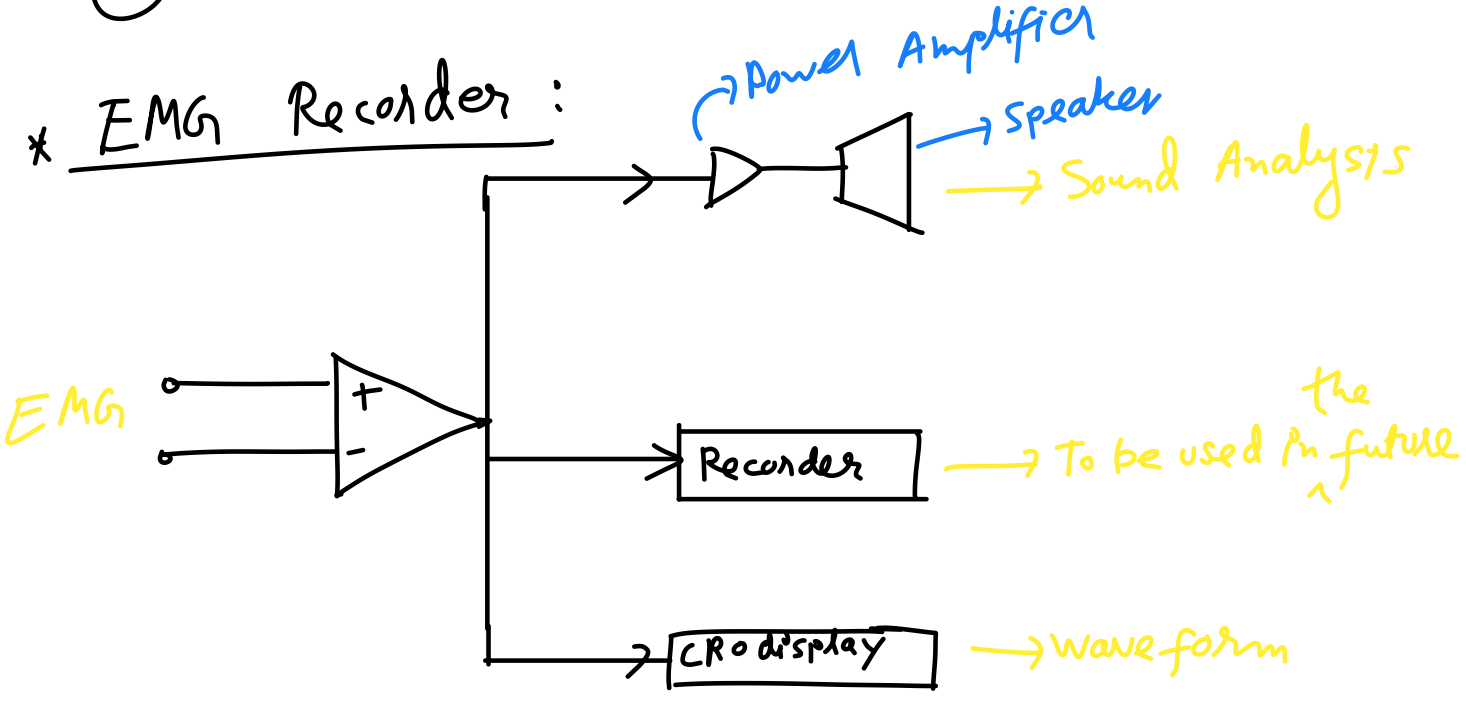
* EMG — EMG is defined as signals generated due to depolarization and repolarization of muscles
  - depolarization → contraction
  - repolarization → relaxation
  - Amplitude → μV (20 - 5000μV)
  - Frequency → 5 Hz - 2KHz

* Acquisition of EMG:

① Surface electrode or patch electrode. (Diameter : 0.5 - 2.5 cm)

② Needle or Wire (Invasive method)

* EMG Recorder:



* Applications of EMG:

① Myopathy: Diseases of muscles can be detected using Myopathy.

② Neuropathy: Problems of stimulations of muscles through Neurons

Conduction Velocity



$$\frac{i_1 - i_2}{t_2 - t_1} = \frac{A}{m\,sec}$$

milli-second.

lec-8

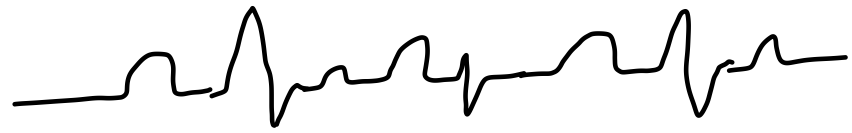
* Artefacts in Biomedical Signals

Artefacts are defined as noises introduced in Biomedical signals during acquisition process.

For e.g.: * ECG signal:

① Artefacts :- ① High freq. EM noise (due to M.S or M.E)

② Motion Artefact: (due to movement)
↳ Baseline Wandering.



Baseline wandering

Isoelectric line



Low frequency Noise.

To remove L.F noise H.P.F is req^{d}

Main Supply line

(III) Powerline Interference

Frequency = 50Hz

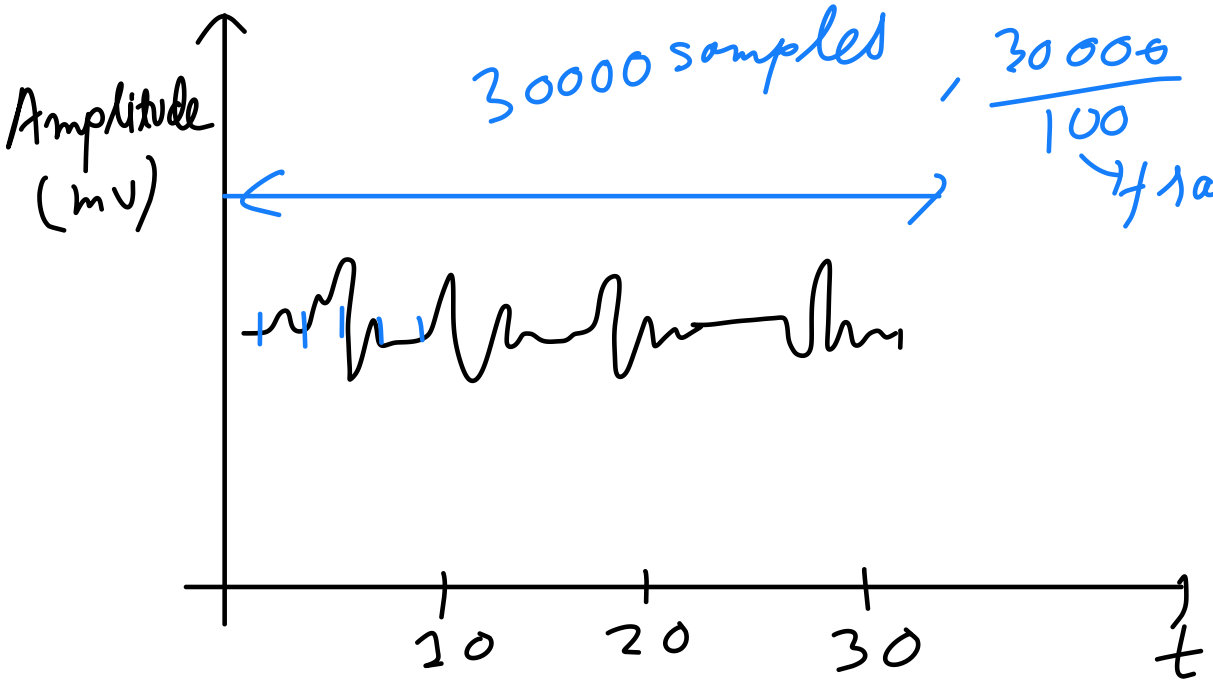
- Def^{n}:

(IV) Maternal interference in Fetal ECG

Mother's ECG signal interfered with Fetus' ECG signal.

* Artefacts Removal Techniques:

① Time domain filtering:



Ex. Synchronized Averaging Technique

$$y_k(m) = x_k(m) + n_k(m), \quad m = 1, 2, \ldots, M \quad (1, 2, \ldots, 100)$$

$x_k(m)$: Signal, $n_k(m)$: Noise

$$\sum_{k=1}^{N} y_k(m) = \sum_{k=1}^{N} x_k(m) + \sum_{k=1}^{N} n_k(m)$$

$\sum_{k=1}^{N} n_k(m) \rightarrow (+ve) + (-ve) = 0$

$$\sum_{K=2}^{N} Y_K(m) = N.x_K(m) \qquad N = 300$$

$$\Rightarrow x_K(m) = \frac{\sum_{K=2}^{N} Y_K(m)}{N}$$

Advantages: ① It is a faster method

② Spectral information of noise is not required

Disadvantages: ① Lack of control parameter in the Filtering technique

Steps:

① Find total no. of samples.

② Find no. of frames

③ Add (summation of frames)

④ Divide summation by N.

Q. $f_s = 1000\text{Hz}$

$t = 30\text{s}.$

$1000 \times 30 = 30 \times 10^3 = 30{,}000$ samples.

Lec-9 (31/08/26)

Moving Average Filtering

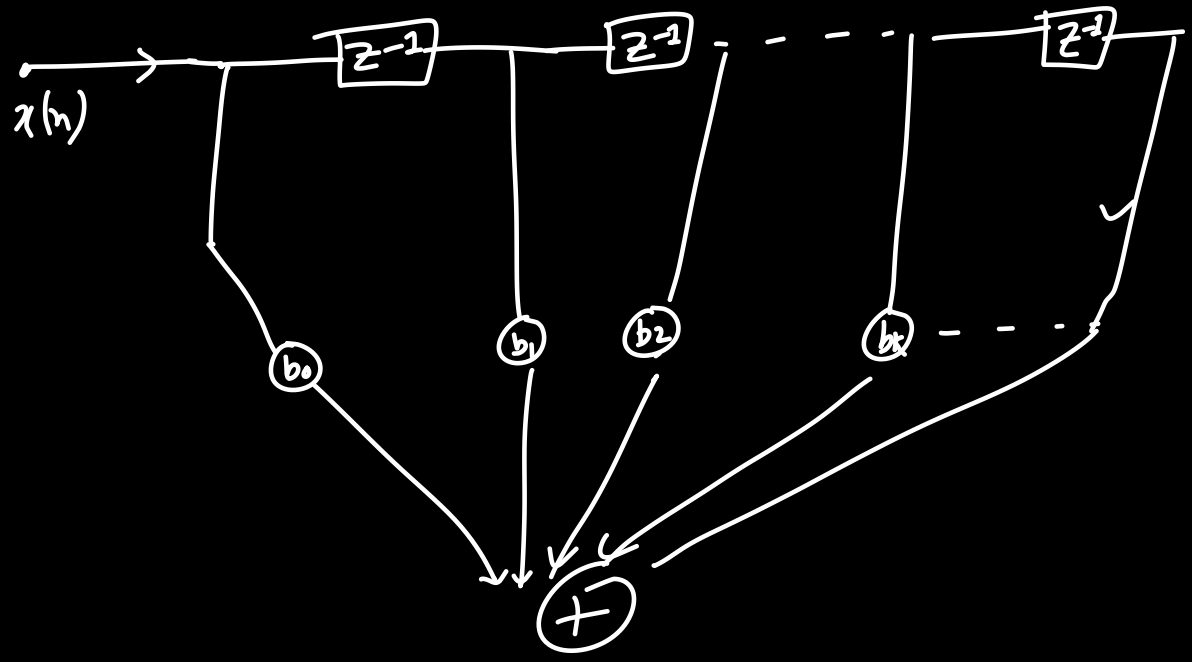
* It is most suitable for Real time application
  ↳ due to less computational time

$$O/P,\ y(n) = \sum_{k=0}^{N} b_k . x(n-k)$$

$$\Rightarrow Y(z) = \sum_{k=0}^{N} b_k . z^{-k} . X(z)$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \sum_{k=0}^{N} b_k . z^{-k}$$

$$\therefore H(z) = \sum_{k=0}^{N} b_k . z^{-k}$$



* It can work on a single i/p of a signal and pt doesnot need multiple realization and that is why pt is more advantageous than Synchronous Averaging Filter.

* Characteristics of Moving Average Filter

(i) · It has finite number of terms
(ii) · It is implemented using an FIR realization which is non-recursive.
(iii) · Output depends on present and past samples
(iv) · Transfer function has no poles except at $z=0$. It provides Stability
(v) Since, o/p depends on present and past few samples, it is causal in nature

vii) It has Linear phase.

Example:

① * Hanning Filter

$$y(n) = \frac{1}{4}\left[x(n) + 2x(n-1) + x(n-2)\right]$$

Q. Find out the Transfer function of the Hanning filter.

Sol^n;

$$H(z) = \sum_{k=0}^{2} b_k \cdot z^{-k} \qquad b_0 = \frac{1}{4},\ b_1 = \frac{1}{2},\ b_2 = \frac{1}{4}$$

Taking z-transform.

$$\Rightarrow Y(z) = \frac{1}{4}\left[z^0 \cdot X(z) + 2z^{-1}X(z) + z^{-2}X(z)\right]$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{1}{4}z^0 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}$$

$$H(z) = \frac{1}{4}\cdot z^0 + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}$$

$$H(z) = \frac{1}{4} + \frac{1}{2}z^{-1} + \frac{1}{4}z^{-2}$$

$$\boxed{H(z) = \frac{1}{4}\left[1 + 2z^{-1} + z^{-2}\right]}$$

$$X(z) = \sum_{n=-\infty}^{+\infty} x(n) \cdot e^{-j\omega n}$$

Q. Find the magnitude and phase response of the Hanning Filter. (Assignment)

z domain $\longrightarrow$ Fourier domain

$re^{j\omega}$

$\hookrightarrow r = 1$

Soln: $H(z) = \frac{1}{4}\left[1 + 2z^{-1} + z^{-2}\right]$

substitute, $z = e^{j\omega T}$

where, $T = 1$

$\Rightarrow z = e^{j\omega}$

$$H(e^{j\omega}) = \frac{1}{4}\left[1 + 2e^{-j\omega} + e^{-2j\omega}\right]$$

$$H(e^{j\omega}) = \frac{1}{4} e^{-j\omega}\left[e^{+j\omega} + 2 + e^{-j\omega}\right]$$

$$\left(\because e^{+j\omega} + e^{-j\omega} = 2\cos(\omega)\right)$$

$$H(e^{j\omega}) = \frac{1}{4} e^{-j\omega}\left[2\cos(\omega) + 2\right]$$

$$H(e^{j\omega}) = \frac{2}{4} e^{-j\omega} \left[\cos(\omega) + 1 \right]$$

$$\left(\because \; 1 + \cos\theta = 2\cos^2\left(\frac{\theta}{2}\right) \right)$$

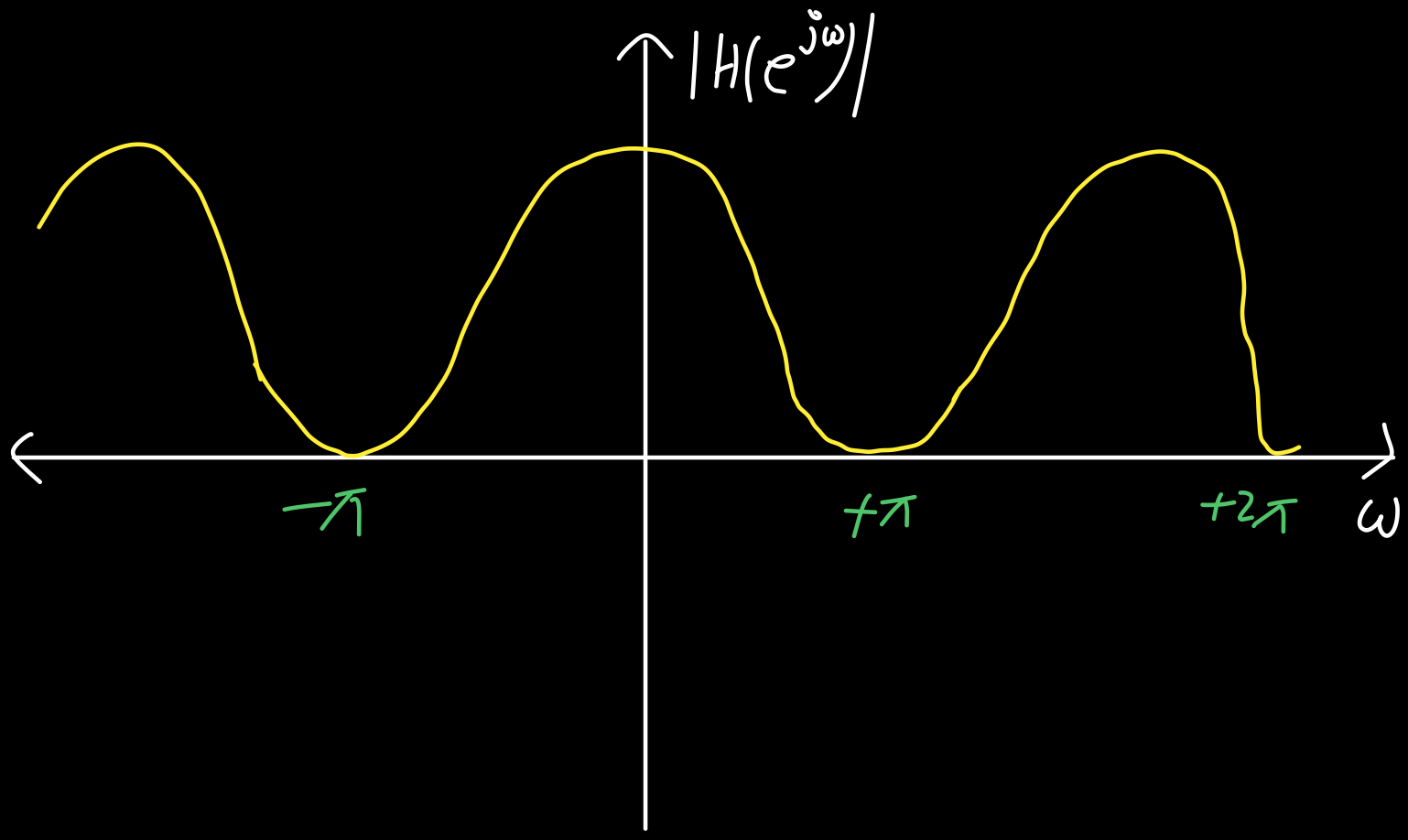
$$H(e^{j\omega}) = \frac{1}{2} e^{-j\omega} \cdot 2\cos^2\left(\frac{\omega}{2}\right)$$

$$\boxed{H(e^{j\omega}) = e^{-j\omega} \cdot \cos^2\left(\frac{\omega}{2}\right)}$$

Magnitude Response:

$$|H(e^{j\omega})| = \left| e^{-j\omega} \cdot \cos^2\left(\frac{\omega}{2}\right) \right|$$

$$|H(e^{j\omega})| = \cos^2\left(\frac{\omega}{2}\right) \quad \text{OR} = \frac{1}{2}[1+\cos\omega]$$



$$\angle H(e^{j\omega}) = \angle\left(e^{-j\omega} \cdot \cos^2\frac{\omega}{2}\right)$$

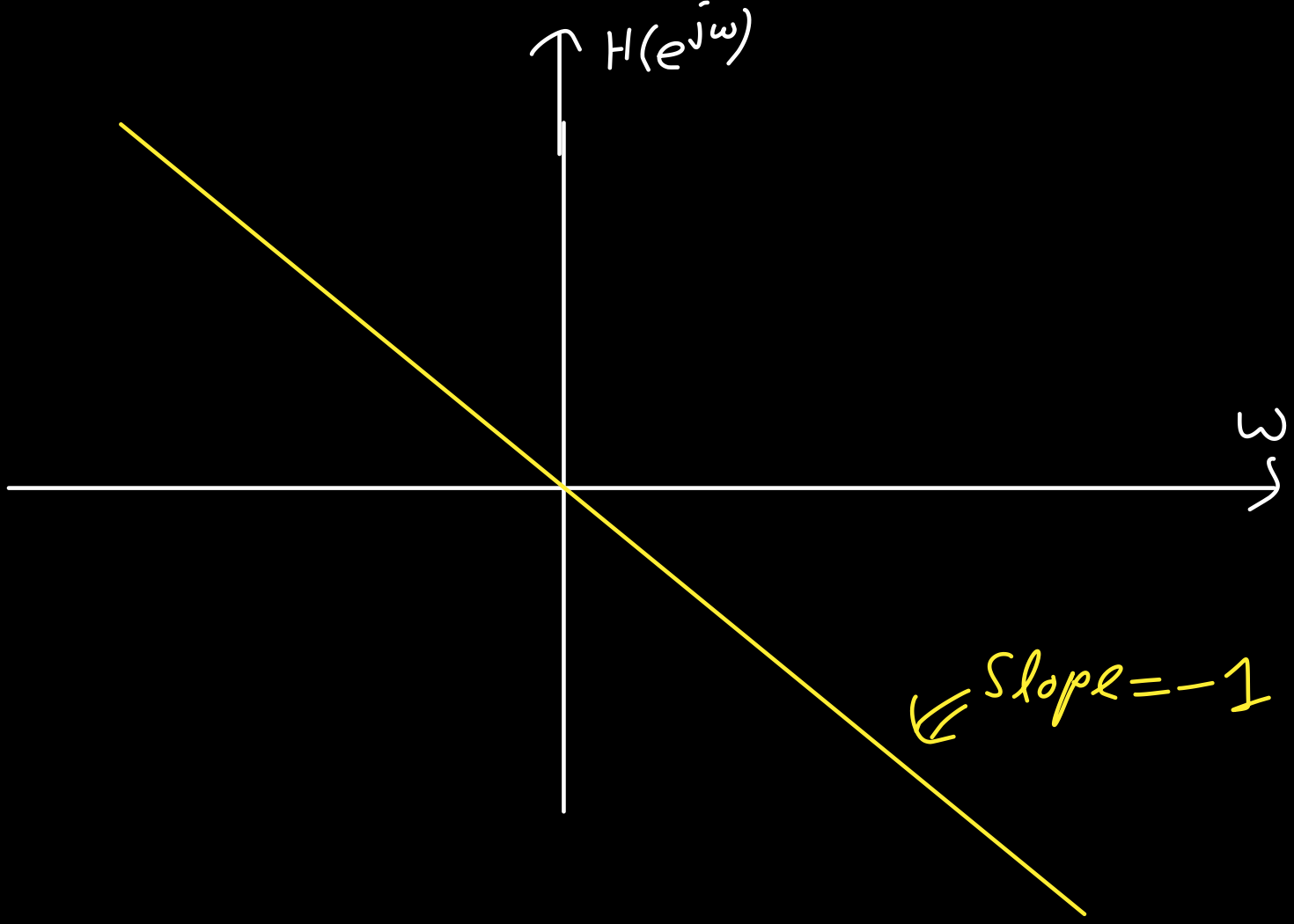
$$= \angle e^{-j\omega} + \angle \cos^2\frac{\omega}{2}$$

$\angle e^{-j\omega}$ ↙ Conjugate part (Phase part)

$\angle \cos^2\frac{\omega}{2}$ ↙ Real part

$$\angle H(e^{j\omega}) = -\omega + 0$$

$$\angle H(e^{j\omega}) = -\omega$$


$H(e^{j\omega})$
ω
Slope = −1

Q. Prove that a Hanning filter has a linear phase.

Soln

$$H(e^{j\omega}) = \cos^2\left(\frac{\omega}{2}\right) \cdot e^{-j\omega}$$

$$\angle H(e^{j\omega}) = \angle\left(e^{-j\omega} \cdot \cos^2\frac{\omega}{2}\right)$$

$$= \angle e^{-j\omega} + \angle \cos^2\frac{\omega}{2}$$

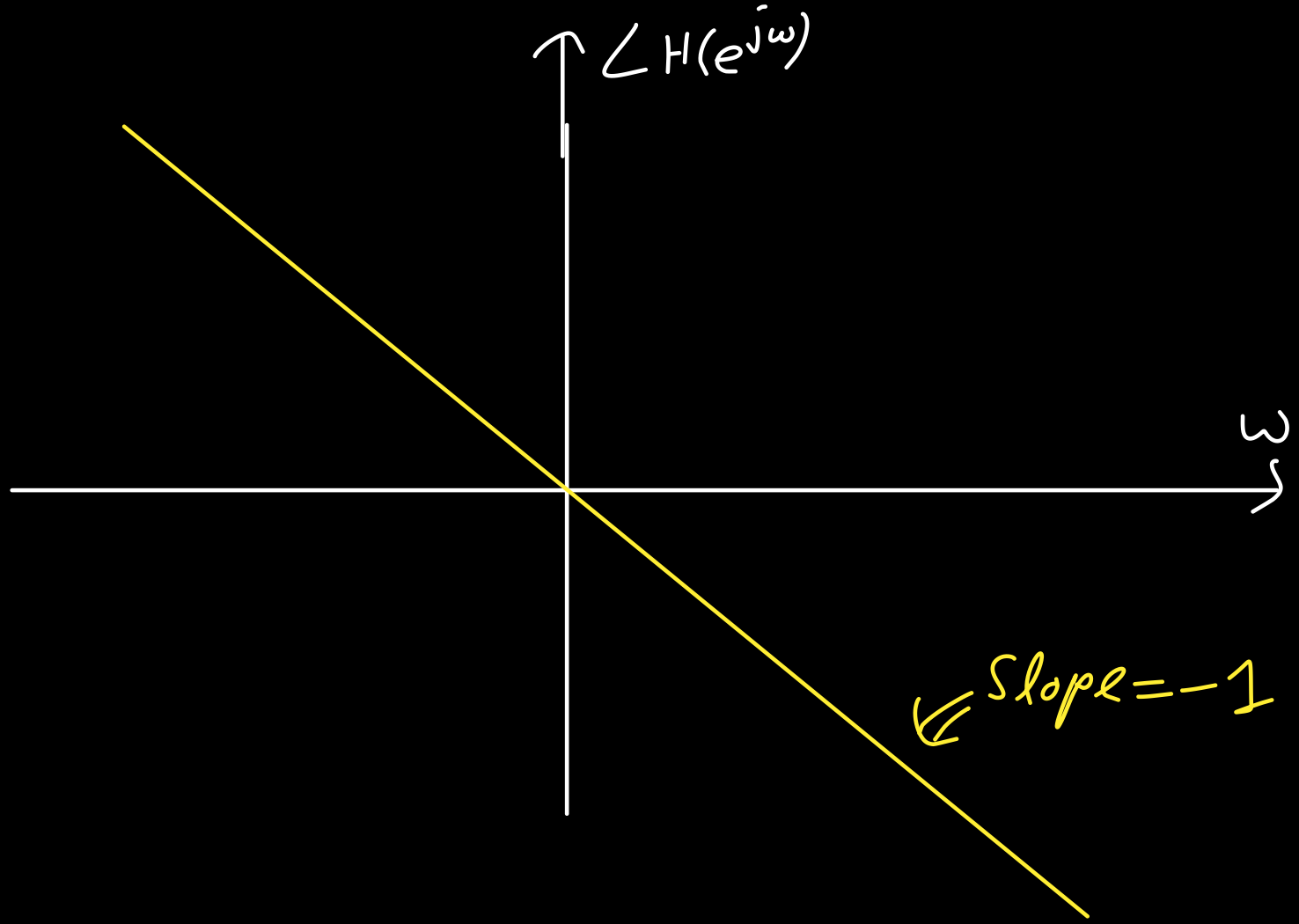
↓ Conjugate part (Phase part)

↓ Real part

$$\angle H(e^{j\omega}) = -\omega + 0$$

$$\angle H(e^{j\omega}) = -\omega$$

↳ Slope = -1

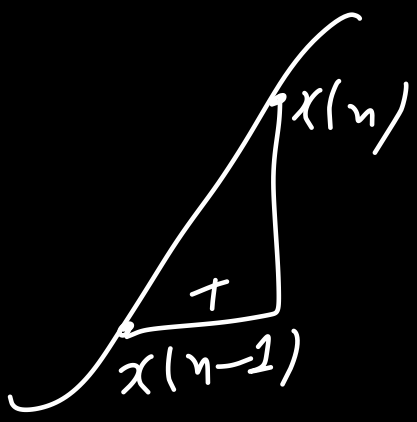

$\angle H(e^{j\omega})$
ω
Slope = −1

Lec-10 [02/09/2026]

* Derivative based approach for Low-frequency Artefacts Removal

$$y(n) = \frac{1}{T}\left[x(n) - x(n-1)\right]$$

T → Sampling Time.



$$\frac{\Delta Y}{\Delta x} = \frac{x(n) - x(n-1)}{T}$$

Q.1 Determine the mag. & phase response of a derivative based filter for low-frequency Artefacts Removal.

Q.2 Prove that the derivative based filter has a linear phase

Q.] Determine the mag. & phase response of a derivative based filter for low-frequency Artefacts removal.

Soln:

$$y(n) = \frac{1}{T}\left[x(n) - x(n-1)\right]$$

Taking Z-transform

$$\Rightarrow Y(z) = \frac{1}{T}\left[X(z) - z^{-1}.X(z)\right]$$

$$\Rightarrow Y(z) = \frac{X(z)}{T}\left[1 - z^{-1}\right]$$

$$\Rightarrow \frac{Y(z)}{X(z)} = \frac{1}{T}\left[1 - z^{-1}\right]$$

$$\Rightarrow H(z) = \frac{1}{T}\left[1 - z^{-1}\right]$$

Putting $z = e^{\pm j\omega}$

$$\Rightarrow H(e^{j\omega}) = \frac{1}{T}\left[1 - e^{-j\omega}\right]$$

Taking $e^{-j\omega/2}$ common

$$\Rightarrow H(e^{j\omega}) = \frac{e^{-j\omega/2}}{T}\left[e^{+\frac{j\omega}{2}} - e^{-\frac{j\omega}{2}}\right]$$

$$\Rightarrow H(e^{j\omega}) = \frac{e^{-\frac{j\omega}{2}}}{T}\left[\left(\cos\frac{\omega}{2} + j\sin\frac{2\omega}{2}\right) - \left(\cos\frac{\omega}{2} - j\sin\frac{2\omega}{2}\right)\right]$$

$$\Rightarrow H(e^{j\omega}) = \frac{e^{-j\omega/2}}{T} \cdot 2j\sin\frac{\omega}{2}$$

Expressing j in polar form as $e^{j\pi/2}$

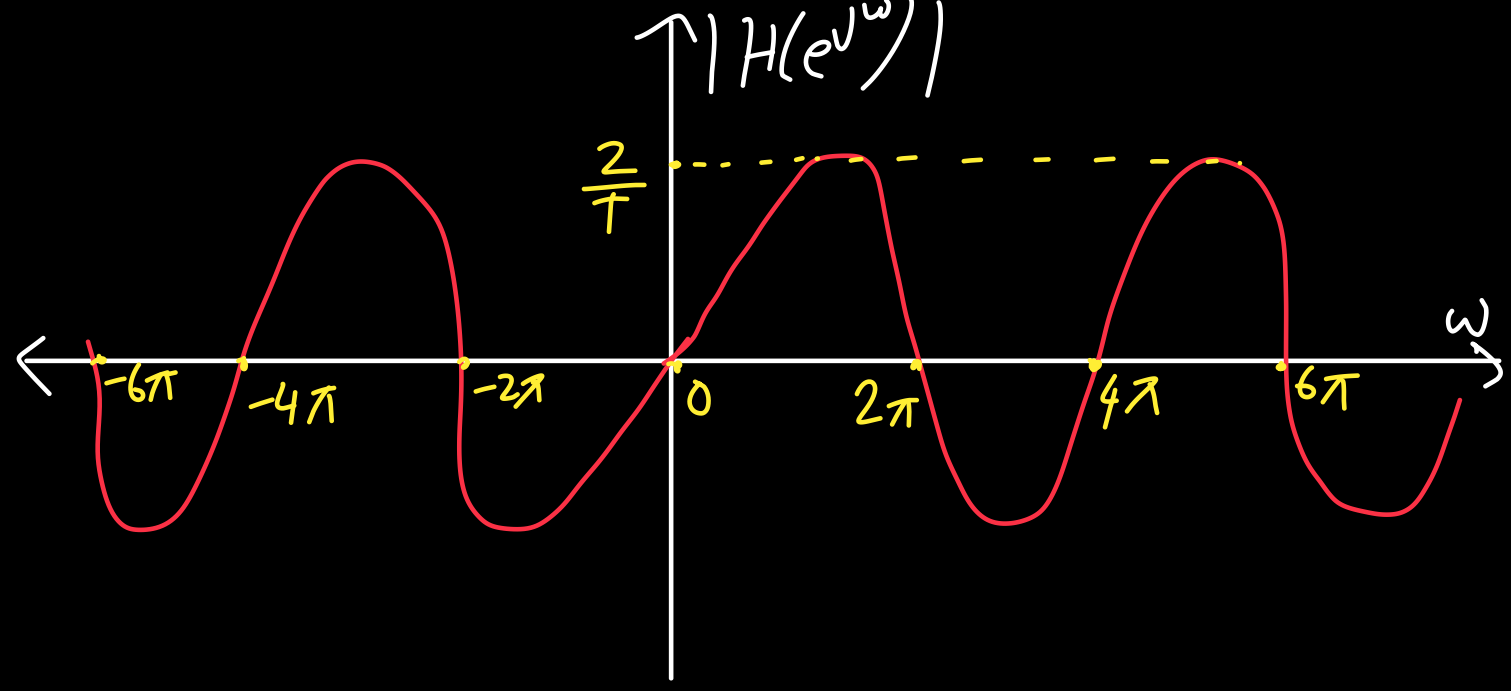
$\left(\because e^{\frac{+j\pi}{2}} = \cos\frac{\pi}{2} + j\sin\frac{\pi}{2} = j\right)$

$$\Rightarrow H(e^{jw}) = \frac{2}{T}\sin\frac{w}{2} \cdot e^{\frac{+j\pi}{2}} \cdot e^{\frac{-jw}{2}}$$

$$\Rightarrow \boxed{H(e^{jw}) = \frac{2}{T}\sin\frac{w}{2}\, e^{+j\left(\frac{\pi}{2} - \frac{w}{2}\right)}}$$

Magnitude:

$$|H(e^{jw})| = \frac{2}{T}\sin\left(\frac{w}{2}\right)$$



Phase: $\angle H(e^{j\omega}) = \angle\left(\frac{2}{T}\sin\frac{\omega}{2}e^{+j\left(\frac{\pi}{2}-\frac{\omega}{2}\right)}\right)$

$$\angle H(e^{j\omega}) = \frac{\pi}{2} - \frac{\omega}{2}$$

